CommuniServe SA – Part 3 Final Report

OPSC6312 Portfolio of Evidence
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Repository: https://github.com/PreciousMoloi/CommuniServeSA_Part3_POE
Hosted API: https://communiservesa.onrender.com/
Demo video: https://drive.google.com/file/d/18ZCTxWemuMIgkK7kuQ48zESAB-PWvqfS/view?usp=drivesdk

1. Purpose

CommuniServe SA lets South Africans discover local services and events, add community listings, and keep using the app offline. It is an Android (Kotlin) app with a Node.js/Express API.

2. Architecture

- Android single-Activity app, programmatic UI, Retrofit/OkHttp, Room offline cache, osmdroid map.
- Firebase Authentication (Google SSO through Credential Manager) and Firebase Cloud Messaging.
- Node.js/Express API: bcrypt password hashing, JWT sessions, Firebase ID-token verification for SSO, JSON-file persistence.
- Hosted on Render (free tier); GitHub Actions runs API tests, Android unit tests and a debug build.

3. Features

- Register, sign in, Google SSO, remember me, guest mode.
- Unified search of services and events; category chips with near-me Google Maps shortcuts.
- User-defined feature 1: Nearby map with service and event pins.
- User-defined feature 2: Community listings with moderation (pending → approved).
- Offline caching and a queue for reviews and listings that syncs on reconnect.
- Push notifications, three languages (English, Afrikaans, isiZulu), dark mode and radius settings.

3A. Changes since Part 2

Part 2 was the v2.0 prototype. Version 3.0 (this submission) adds the following.

Area	Part 2 (v2.0)	Part 3 (v3.0)
Sign-in	Email and password only	Google single sign-on through Firebase; the API verifies the Firebase ID token (signature, issuer, audience, expiry); guest mode and remember-me
Notifications	None	Firebase Cloud Messaging push (open, background, closed); device token registered with the API; Android 13+ permission prompt
Offline	Limited saved items	Room cache plus a queue for reviews and listings that syncs automatically on reconnect, with a manual sync button and a waiting-actions indicator
Languages	English	English, Afrikaans and isiZulu on every screen; switch on login and in Settings
User-defined features	Basic listings	Nearby map (OpenStreetMap, pins, radius, live location) and community listings with moderation (pending, then approved)
Search and discovery	Simple list	Unified search of services and events, category chips and near-me Google Maps shortcuts (health, food, transport, repairs, cleaning)
User interface	Basic screens	Redesigned login and registration, bottom navigation, Material icons, profile header, dark mode, edge-to-edge layout, 48 dp touch targets
Error handling	Technical messages	Plain-language messages in three languages, limited-mode card with Retry Connection, friendly 404 and offline messages
API storage	In memory (lost on restart)	JSON-file persistence (server/data/db.json); on Render free tier it still resets

		on redeploy
API security	Basic	bcrypt (cost 12), required JWT secret in production, rate-limited auth routes, closed CORS, stronger validation
Hosting	Local only	Hosted on Render over HTTPS; physical phones use it by default
Testing and CI	None or manual	11 API tests, 31 Android unit tests, GitHub Actions on every push (green), 16 manual device tests
Publishing	Not prepared	Target SDK 36, signed release build, store graphics, privacy policy, listing text and documented Play Console sign-up to the payment step
Documentation	Report only	README, release notes, test evidence with screenshots, AI use write-up, Play Store evidence

4. Screenshots

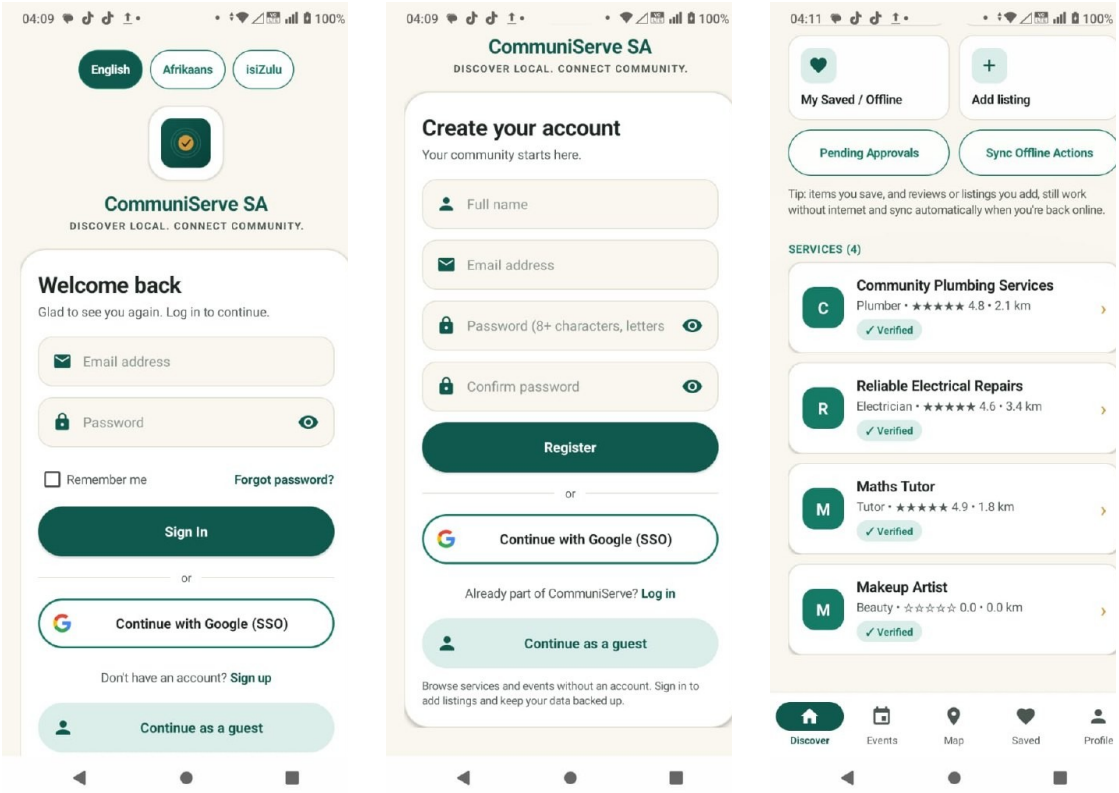


Figure 1: Login, registration and Discover.

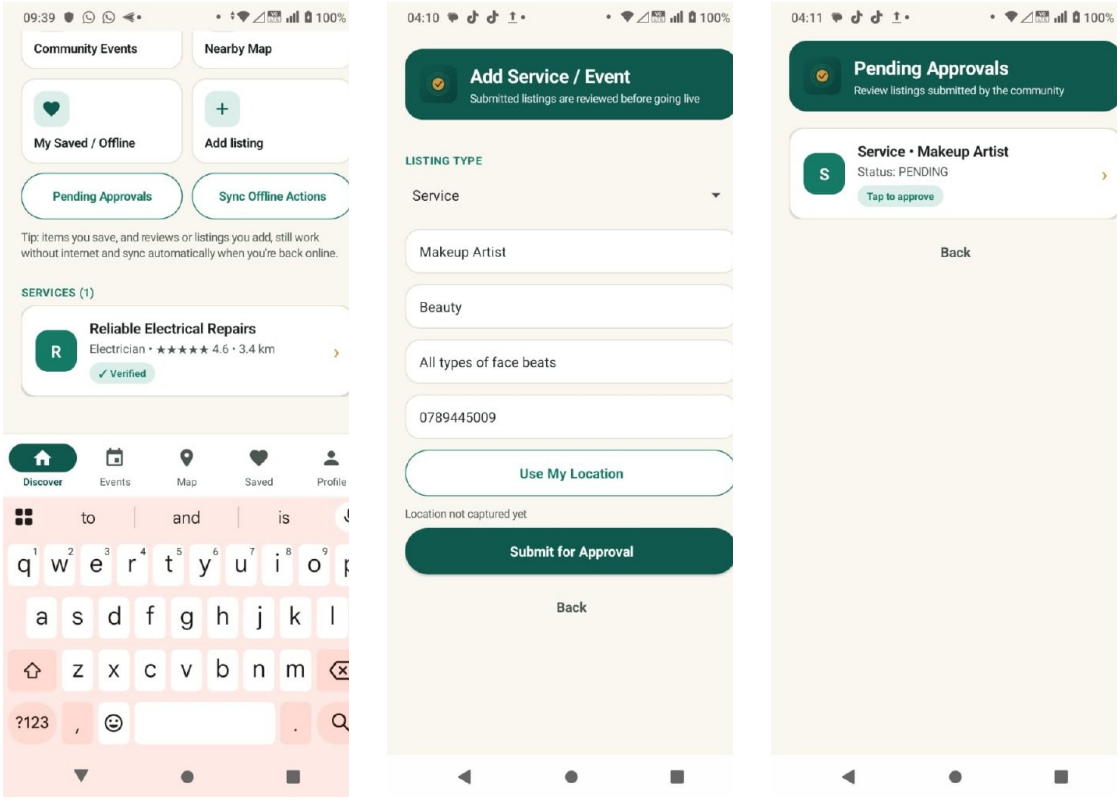


Figure 2: Search, add listing and pending approvals.

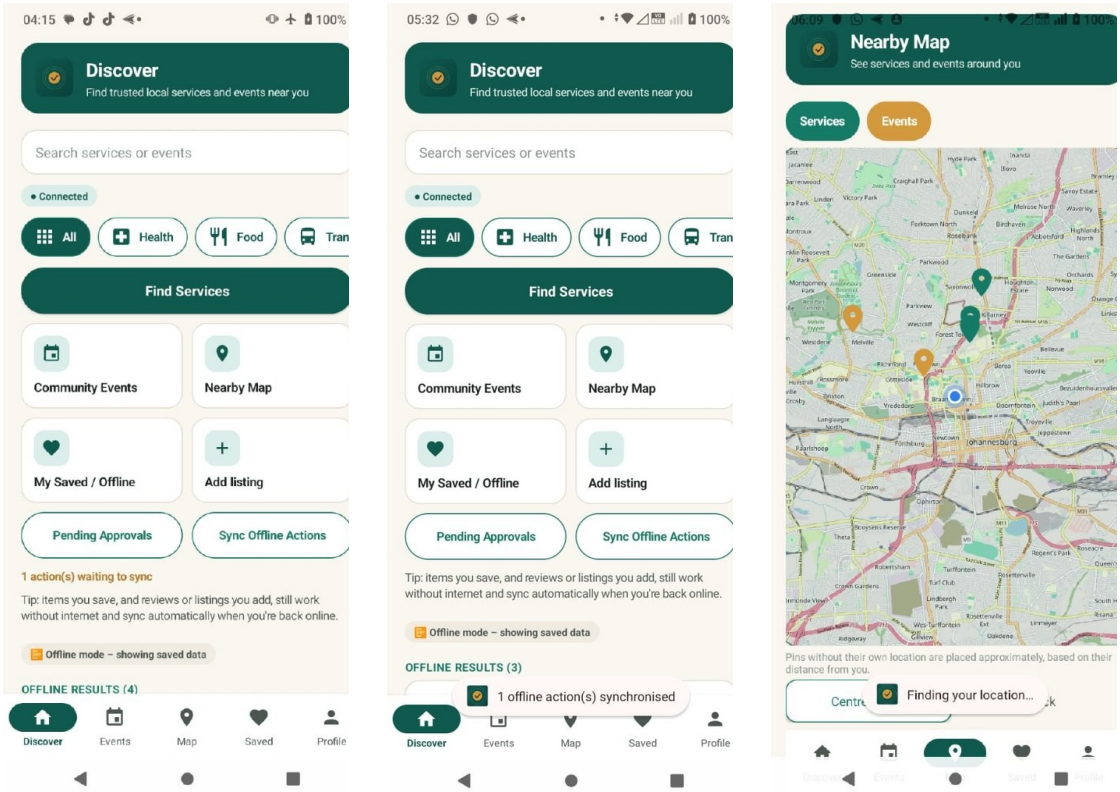


Figure 3: Offline queue, sync success and Nearby map.

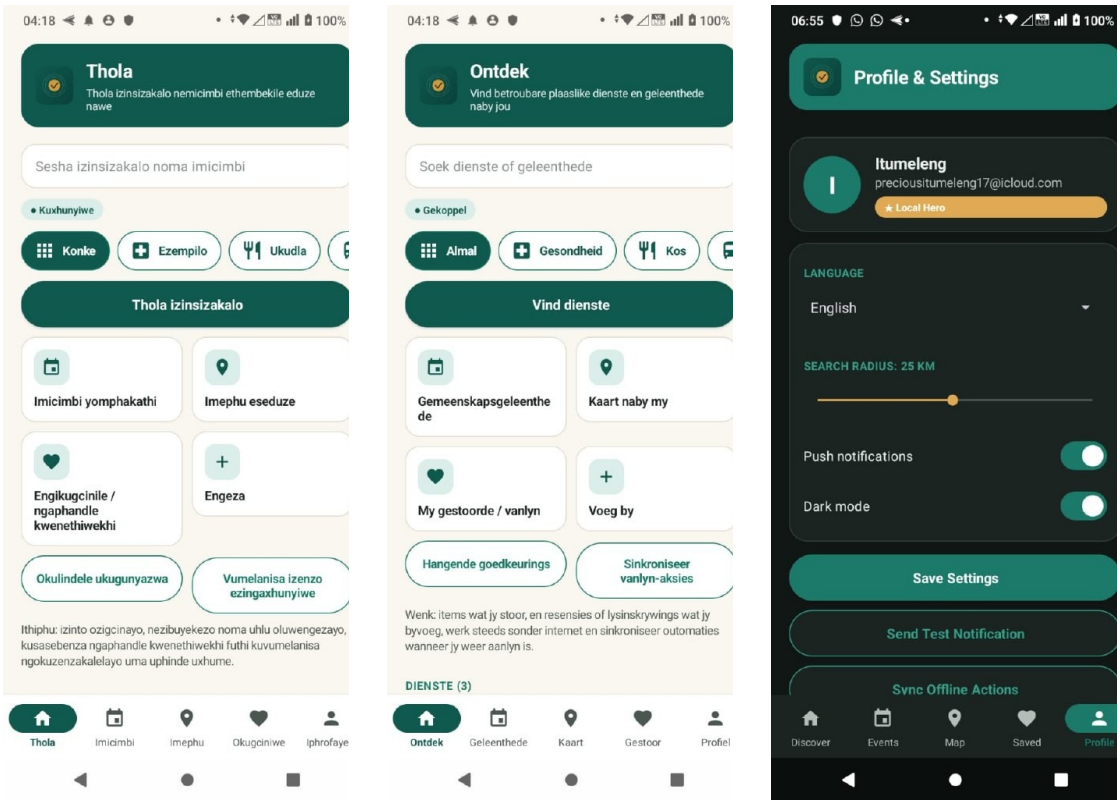


Figure 4: isiZulu, Afrikaans and dark mode.

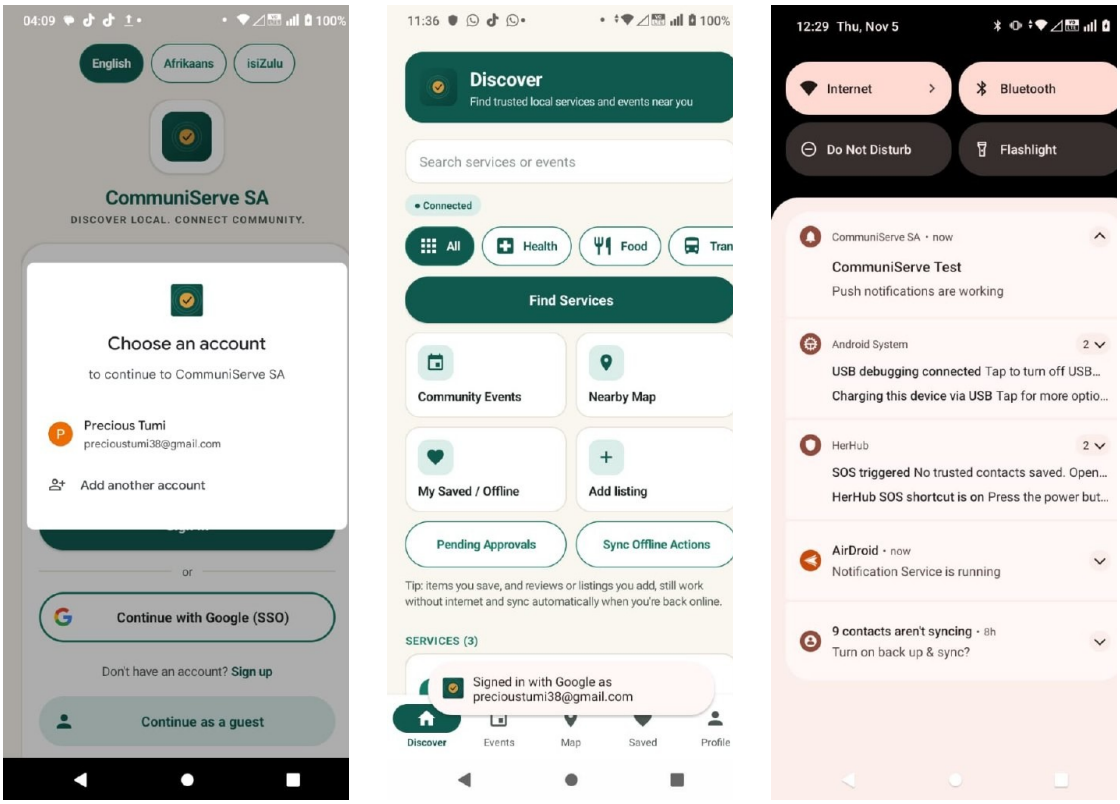


Figure 5: Google sign-in and push notification.

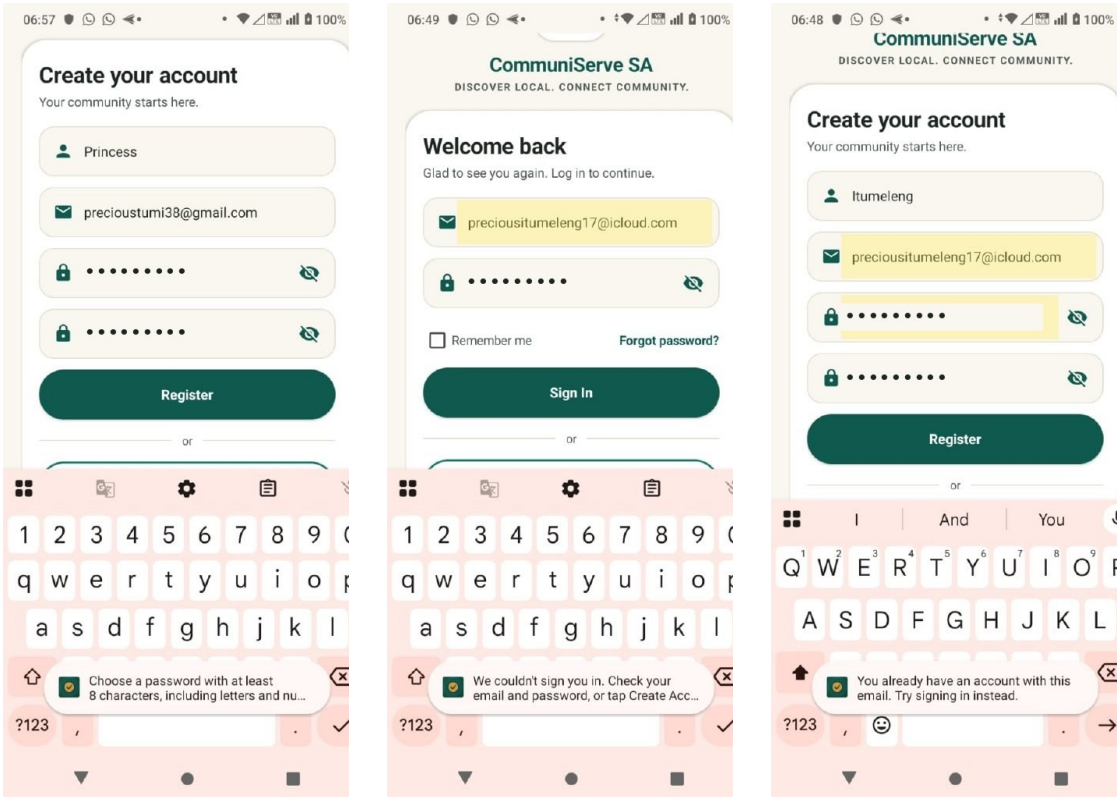


Figure 6: Friendly validation and error messages.

5. Testing

Area	Result
API tests (npm test)	11 of 11 pass
Android unit tests	31 tests, 0 failures
Debug build / release APK	Successful; release APK signed
GitHub Actions	Green on master
Manual tests on a physical phone	15 of 16 pass; TC-09 documented limitation

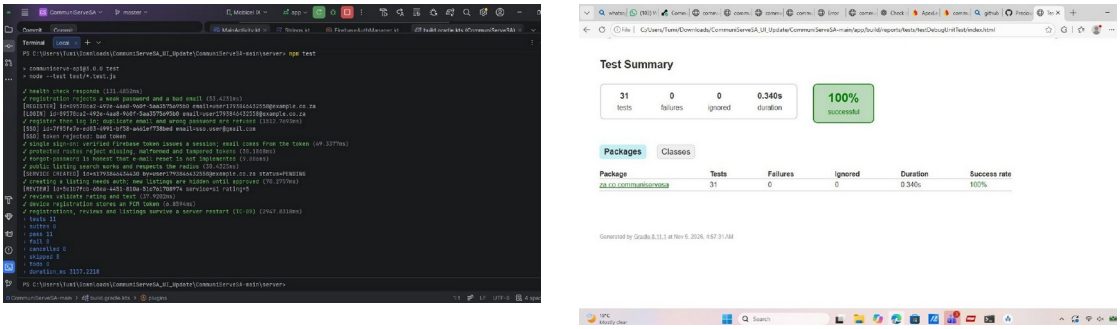


Figure 7: API tests and Android unit-test report.

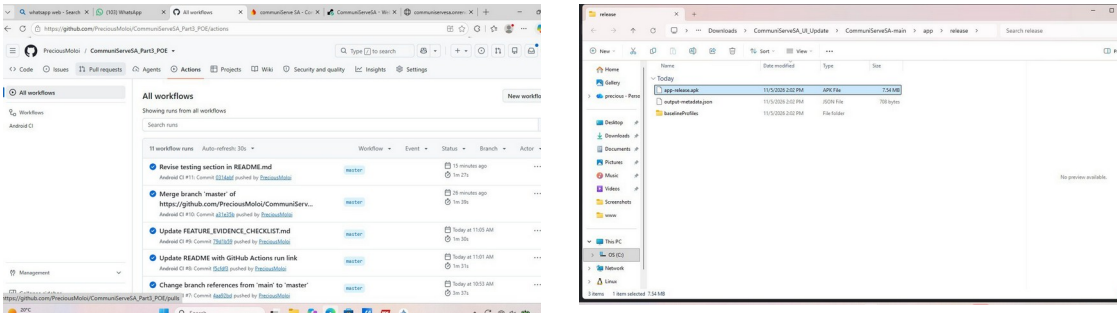


Figure 8: GitHub Actions and signed release APK.

The full test register (AT-01 to AT-08, TC-01 to TC-16) and the defect log are in docs/TEST_EVIDENCE.md.

6. Defects found and fixed

- Phone used the emulator address 10.0.2.2 – hosted HTTPS default added.
- Render ran the old v2.0 API (SSO 404) – redeployed v3.0.
- Wrong default API URL – corrected.
- Windows persistence test failure – test waits for the save.
- Search results hidden below the fold – unified search.

- Missing string key and duplicate drawable folder – fixed.
- Overlapping bottom navigation – fallback insets.

7. Known limitation

TC-09 fails on the hosted API: Render's free tier has an ephemeral disk, so db.json is reset on restart or redeploy. The same behaviour passes locally (AT-05). A persistent disk or database would resolve it.

8. Security and privacy

Passwords are hashed with bcrypt, sessions use JWT with a required secret in production, CORS is closed by default, SSO tokens are verified server-side, and password reset is disclosed as not implemented. The privacy policy is in docs/PRIVACY_POLICY.md.

9. AI use

Claude assisted with UI redesign, error messages, translations, security fixes, connectivity fixes and documentation. We ran and verified everything. See docs/AI_USE_WRITEUP.md.

10. Google Play Store readiness

Publishing on Google Play needs a one-off US$25 developer registration fee. As instructed by our lecturer, we captured each step of the Play Console sign-up up to the payment screen. Personal details are blurred. The signed release APK, icon, feature graphic, privacy policy and listing text are ready. Demo video: https://drive.google.com/file/d/18ZCTxWemuMIgkK7kuQ48zESAB-PWvqfS/view?usp=drivesdk

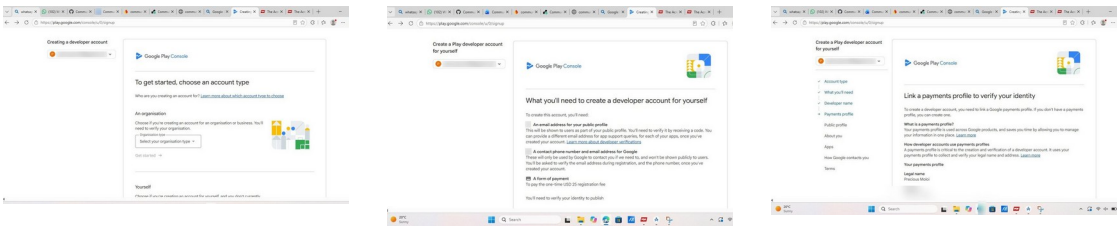


Figure 9: Account type, requirements and payments profile.

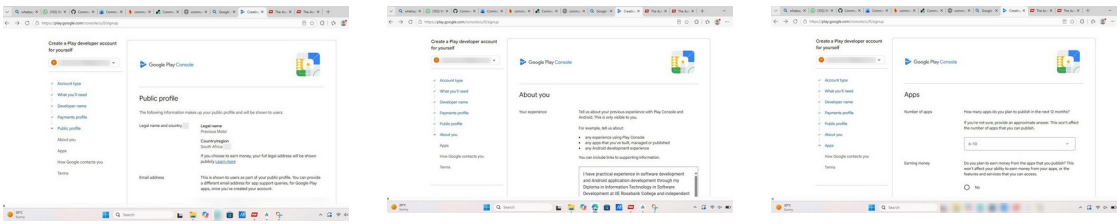


Figure 10: Public profile, about you and apps.

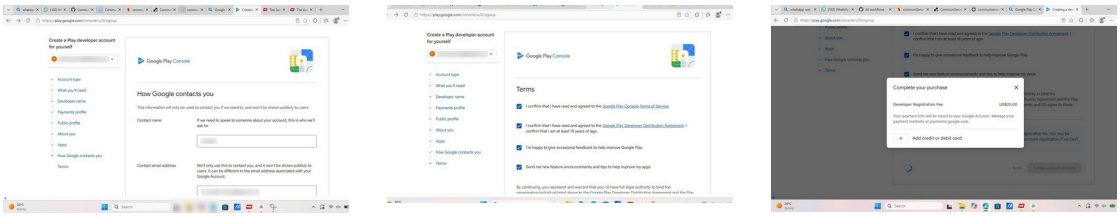


Figure 11: Contact details, terms and the US$25 payment step (stopped here).

All 17 steps are in docs/PLAY_STORE_EVIDENCE.md.